



Small Yet Performant

Efficient Machine Learning: Titanic Survival Study

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PROJECT OBJECTIVE

Goal: Develop a Parsimonious Model.

Core Requirements:

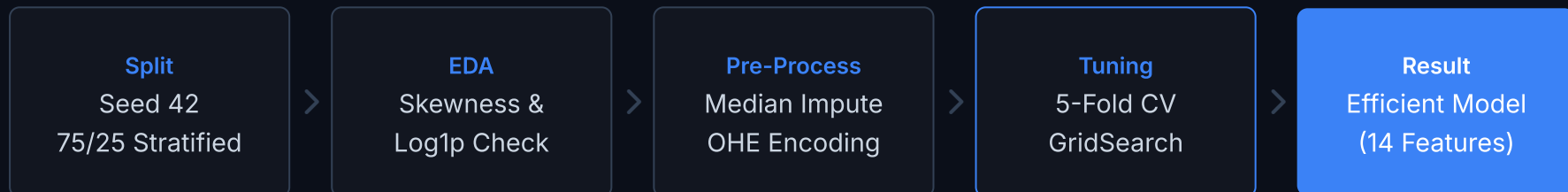
- Balance high predictive power with model simplicity.
- Empirical feature reduction via contribution analysis.
- Strict reproducibility via **Seed 42**.

Efficiency Point

Emphasis is placed on an efficient model that maintains performance while using significantly fewer features (e.g., Strategy C with 14 features).

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PROJECT PIPELINE



"Data-driven feature selection ensures parsimony without sacrificing accuracy."

METHODOLOGY: PREPROCESSING



Data handling choices were justified by **Exploratory Data Analysis**:

- **Skewness:** Right-skewed Fare normalized via $\log_{1p}$ transformation.
- **Missingness:** Median imputation (Numeric) and "Missing" labels (Categorical).
- **Consistency:** Fixed Seed 42 applied to all randomized operations for strict reproducibility.

MODEL 1: K-NEAREST NEIGHBORS

k-NN achieved high predictive performance through scaling and PCA-based dimensionality reduction.

Performance Metrics:

Accuracy (Test): 80.72%

ROC-AUC: 0.8208

Used **PCA (95% variance)** to handle feature explosion from OHE.



RANDOM FOREST COMPLEXITY MATRIX

Strategy A: Large

Maximum info retention. 1,378 Features.
Accuracy: 80.27%.

Strategy B: Eng.

Domain-motivated Interactions. 75
Features. Test Accuracy: 78.03%.

Strategy C: Select

Efficiency Focus. 14 Features. Accuracy:
79.82%.

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THE PARSIMONY FRONTIER

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Features (Selection)

vs. 1,378 (Large Model)

"Dimensional reduction: Trading 1,364 extra features for a 0.45% accuracy gain is inefficient. Strategy C is the strategic choice."

Pruning concentrates the model on strong predictors while ignoring One-Hot noise.

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DISCUSSION: GENERALIZATION GAP

CV Accuracy: 83.38%



Test Accuracy: 78.03%

Why **Strategy B** (Engineering) underperformed:

- **Discretization Loss:** Binning Age/Fare discarded detail trees use natively.
- **Redundancy:** New interaction terms duplicated existing augmented signals.
- **Instability:** Manual features overfitted to partition noise in cross-validation.

PERFORMANCE COMPARISON MATRIX

Strategy	Test Accuracy	ROC-AUC	Features	Project Verdict
k-NN (PCA)	80.72%	0.8208	Reduced	Max Predictive Power
RF: Strategy A	80.27%	-	1,378	Inefficient Info
RF: Strategy C	79.82%	-	14	OPTIMAL CHOICE
Decision Tree	77.58%	-	75	High Variance

BENCHMARK VISUALIZATIONS



Efficiency Goal Met: Strategy C delivers a "Small yet Performant" model.

CONCLUSION: EFFICIENCY VERIFIED



The CorrRFSelect (Strategy C) is the project winner.

Our project confirms that parsimony and effectiveness are not mutually exclusive. By using just 14 features, we achieved production-ready performance with minimal complexity and maximum interpretability.

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Questions & Discussion

